Pragmatic Functional Programming

with TypeScript - Day 2

Olarn U. - Odds Team

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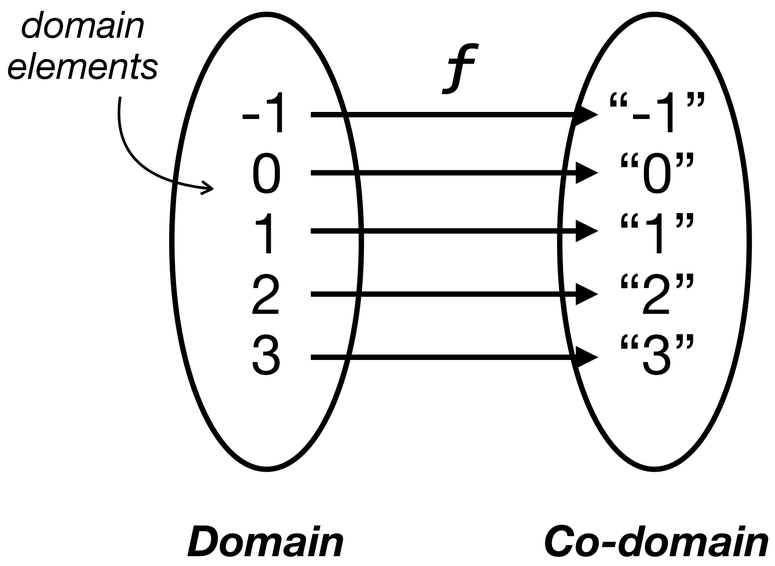
Types

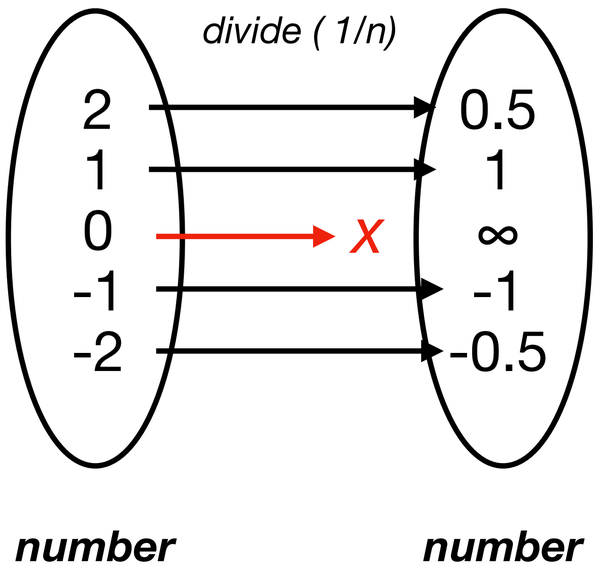
Let the compiler help you
catch bugs early

Domain & Co-domain

A Function is a mapping between all elements of 2 sets.



Total (Pure) Function

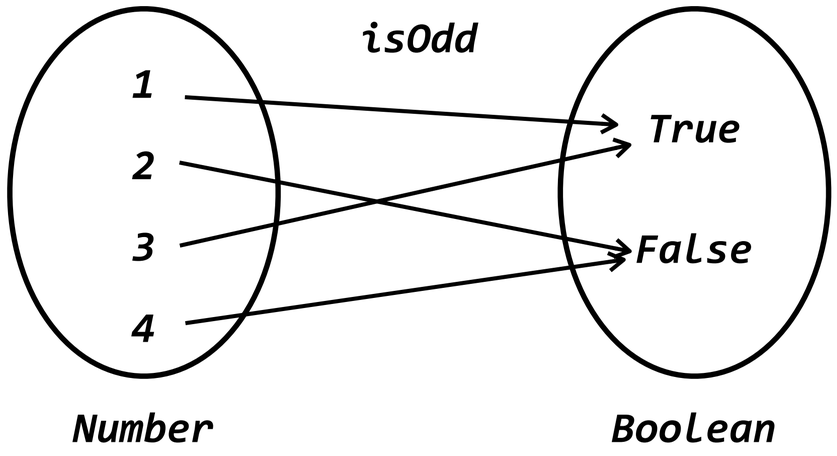


Partial Function

There are more problems about the input type and it's elements

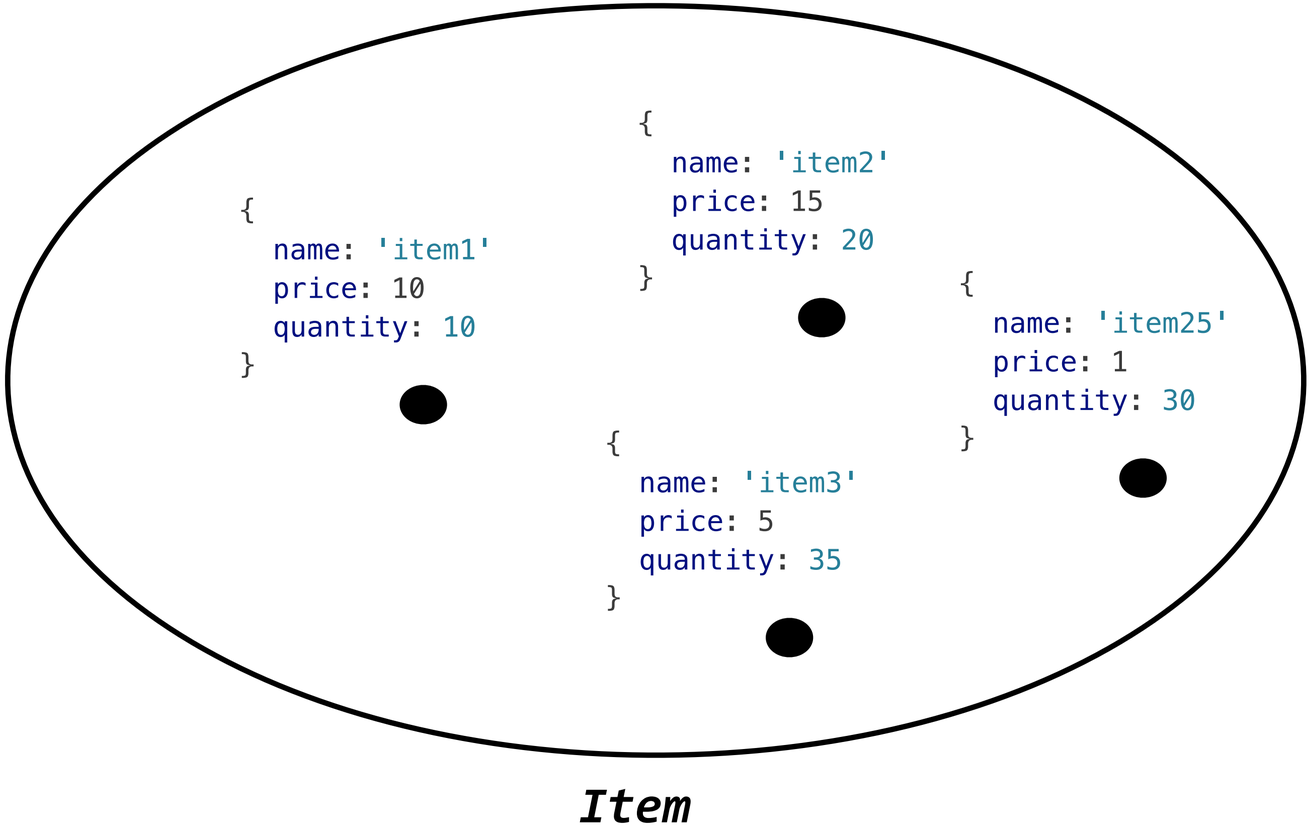
Type and Set

Type focus on it's Class, Set focus on it's elements



Type and Set

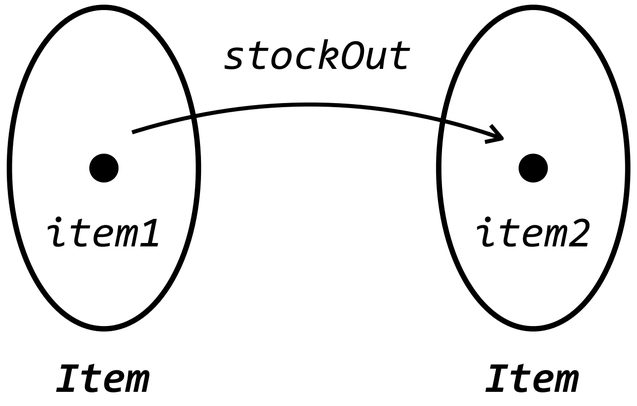
Let's see the objects inside the same set




```
type Item = Readonly<{
  name: string
  price: number
  quantity: number
}>
```

Immutable Object

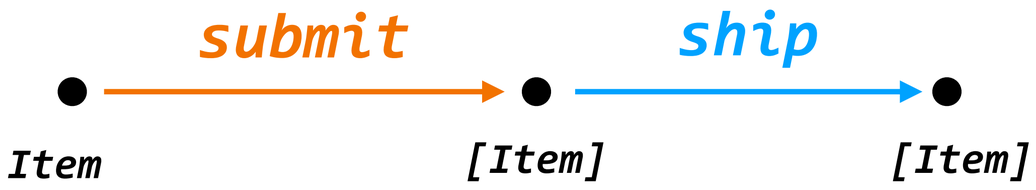
In functional programming, changing any properties of an object results in a new object.




```typescript
type Item = Readonly<{
  name: string
  price: number
  quantity: number
}>

const item1: Item = { name: 'Item 1', price: 10, quantity: 10 }

const stockOut = (qty: number) => (item: Item): Item => ({
  ...item,
  quantity: item.quantity - qty
})

const item2 = stockOut(item1)(5)
// item2 = { name: 'Item 1', price: 20, quantity: 5 }
```

What About List?

Can we ensure that the list is not empty?

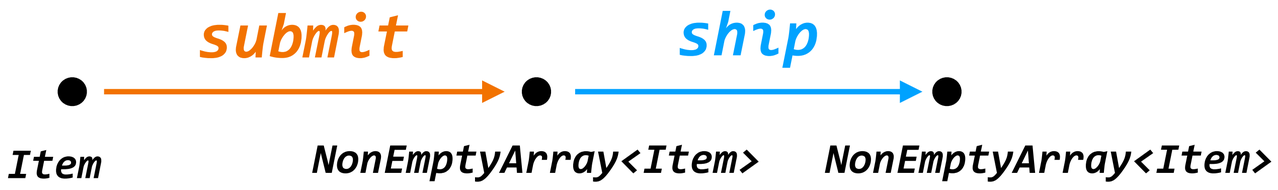



```
type Item = Readonly<{
  name: string
  price: number
  quantity: number
}>

const submit = (items: [Item]): [Item] => {...}
const ship = (items: [Item]): [Item] => {...}
```

- The **`submit`** function accept item and return *list of items*.
- The **`ship`** function accept *list of items*, and return *list of items*. The function require at lease 1 item.
- The question is, how to ensure that the list (for function **ship**) is not an empty-list?

Type Checking

in compile time, not run time.




```
type Item = Readonly<{
  name: string
  price: number
  quantity: number
}>
```

```
const addFee = (item: Item): Item => ({ ...item, price: item.price * 1.01 })

const submit = (items: Item[]) => (item: Item): NonEmptyArray<Item> =>
  isNonEmptyArray(items) ? append(item)(items) : [item]

const ship = (items: NonEmptyArray<Item>): ReadonlyArray<Item> => map(addFee)(items)

const f = flow(submit([]), ship)
```

From Business Requirement

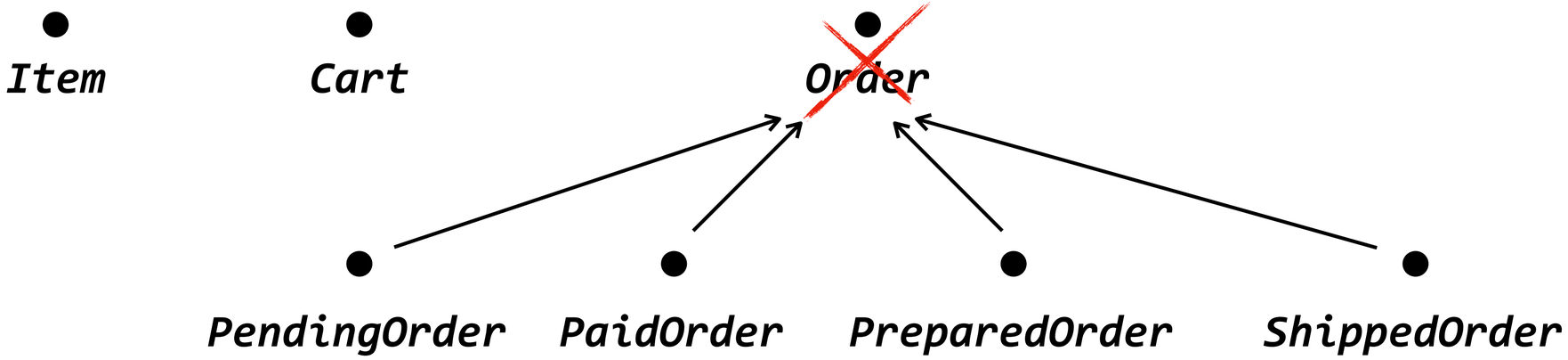
Online shopping

The way we currently doing our business is, we have item(s) that can be added to cart, a cart can be checked out, which create an order, but since no payment is made, it is pending any operation, afterward, orders must be paid, and the paid order would be prepared and, subsequently, shipped.

From Business Requirement

Let's extract nouns as types

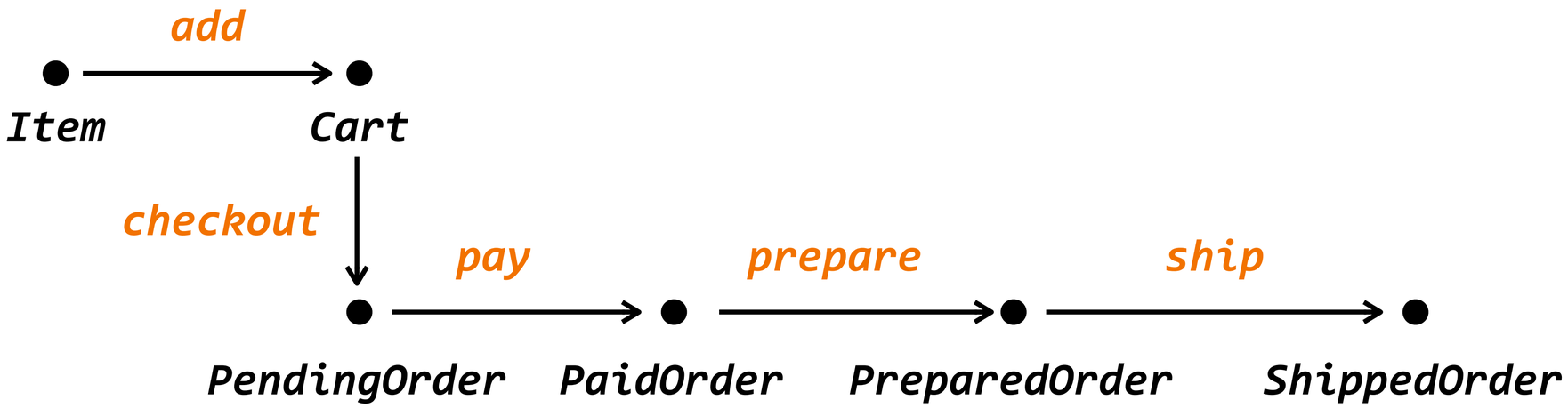
The way we currently doing our business is, we have **item**(s) that can be added to cart, a **cart** can be checked out, which create an **order**, but since no payment is made, it is ***pending*** any operation, afterward, orders must be paid, and the ***paid order*** would be ***prepared*** and, subsequently, ***shipped***.



From Business Requirement

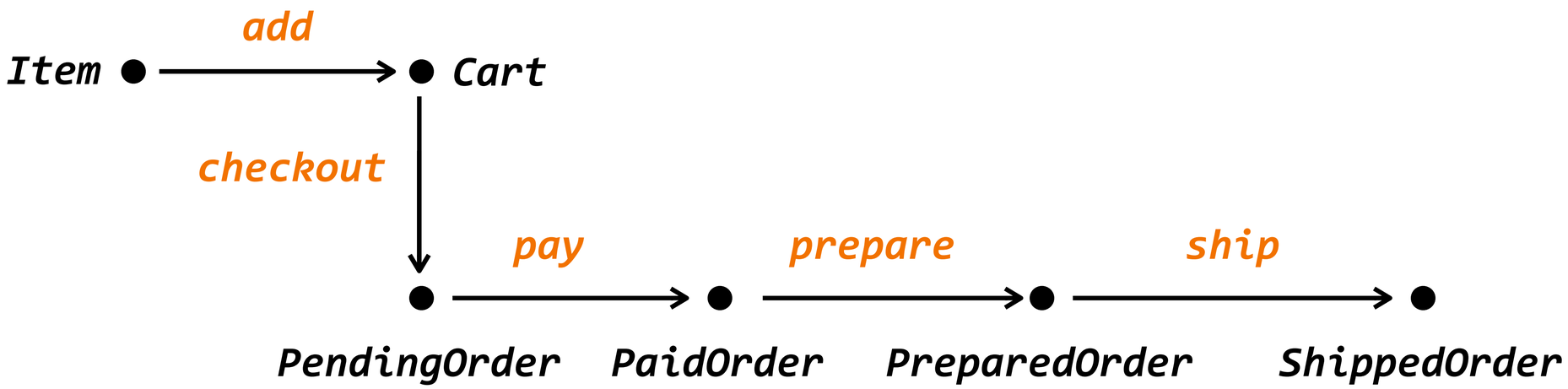
Let's extract verbs as functions

The way we currently doing our business is, we have **item**(s) that can be added to cart, a **cart** can be checked out, which create an **order**, but since no payment is made, it is ***pending*** any operation, afterward, orders must be paid, and the ***paid order*** would be ***prepared*** and, subsequently, ***shipped***.



From Business Requirements

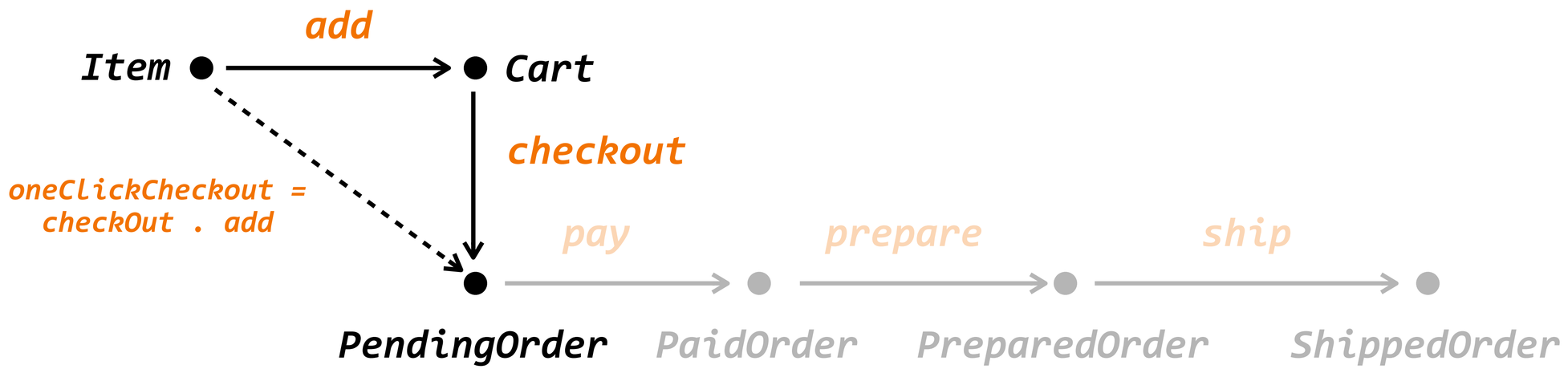
Now, we have functions and types that matched business requirement and the compiler help us to prevent from runtime type mismatch error.




```
const add      = (item:  Item):          Cart
const checkout = (cart:  Cart):          PendingOrder
const pay      = (order: PendingOrder):  PaidOrder
const prepare  = (order: PaidOrder):     PreparedOrder
const ship     = (order: PreparedOrder): ShippedOrder
```

Requirement Changed

Business want to add “1-click checkout” feature




```
const add      = (item:  Item):          Cart
const checkout = (cart:  Cart):          PendingOrder
const pay      = (order: PendingOrder):  PaidOrder
const prepare  = (order: PaidOrder):     PreparedOrder
const ship     = (order: PreparedOrder): ShippedOrder
```

```
const oneClickCheckOut = flow(add, checkout)
```

Summary

- Functional Programming uses ***types*** to model business concepts clearly.
- Types can express stronger constraints, such as **NonEmptyArray**, to ***prevent invalid states***. With compile-time type checking, many errors can be caught before runtime.
- Business requirements can be translated into function composition.
- When functions and types align with the business workflow, the code becomes easier to understand, safer to change, and harder to misuse.
- The compiler becomes a partner that helps us prevent runtime type mismatch errors early.

Option

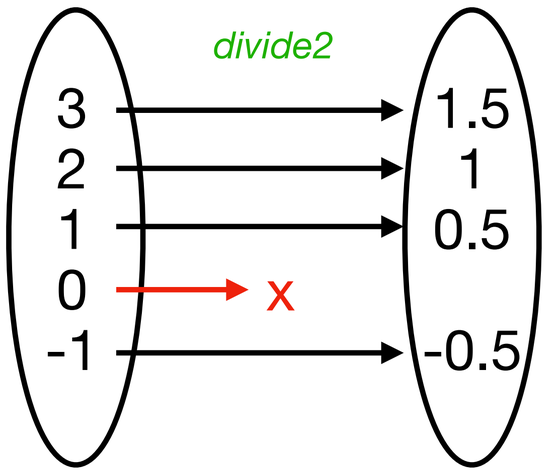
Non-Deterministic Function Handling

Non-Deterministic (Partial) Function

The function that accept only certain of value of the input type universe

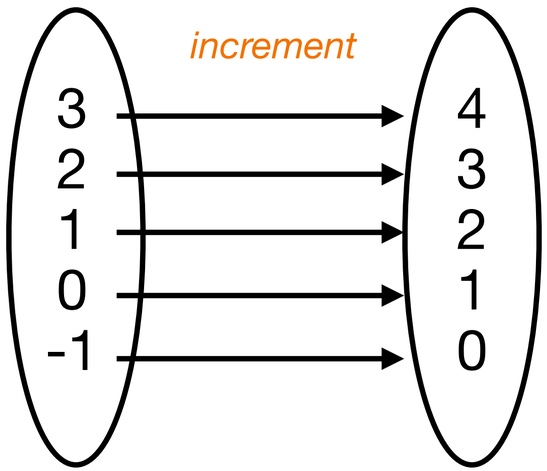
Non-Deterministic Function

```
function divide2(x: number) {
  return 2 / x
}
```



Deterministic Function

```
function increment(x: number) {
  return x + 1
}
```



Function in FP should be total, but how to handle this?

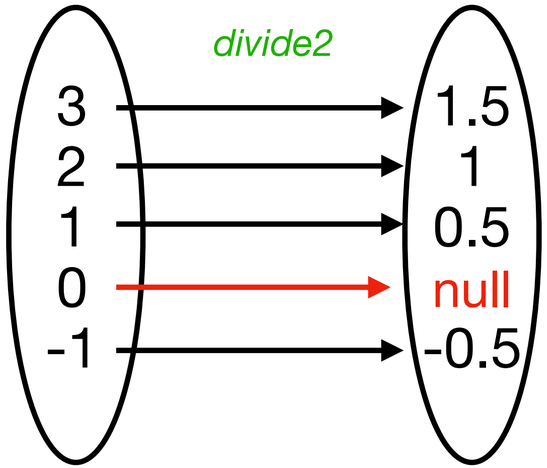
Let's return something indicate output is not available for 0.

Null?

That means we need to consider null (undefined in TS) everywhere possible…

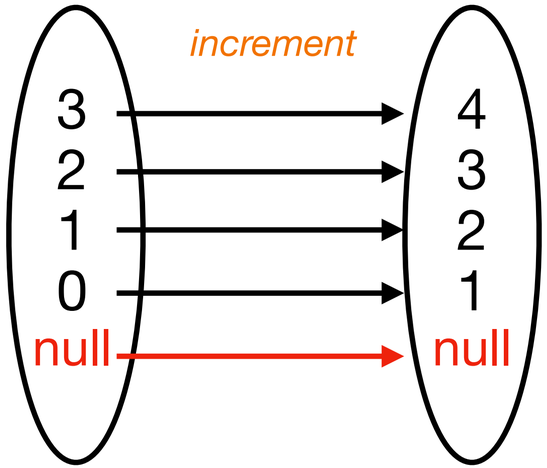
Non-Deterministic Function

```
function divide2(x: number): number | undefined {
  if (x === 0) {
     return undefined
  }
  return 2 / x
}
```



Deterministic Function

```
function increment(x: number): number | undefined {
  if (x < 0) {
     return undefined
  }
  return x + 1
}
```



Missing considering the null value can lead to unexpected behaviour and leads to bug.

Do you ever heard…

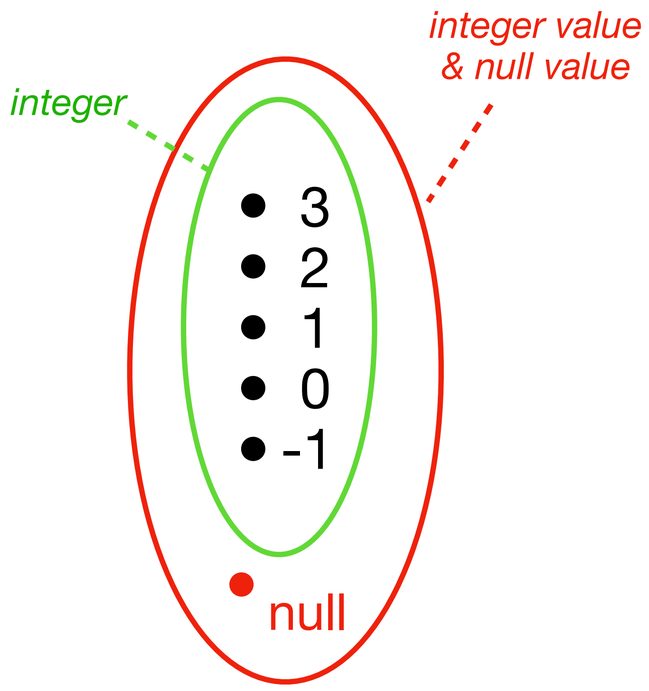
Null is a one billion dollar mistake.

By Tony Hoare who invented `null` in 1965 while designing the ALGOL W programming language.

Null Value

What's wrong with null?

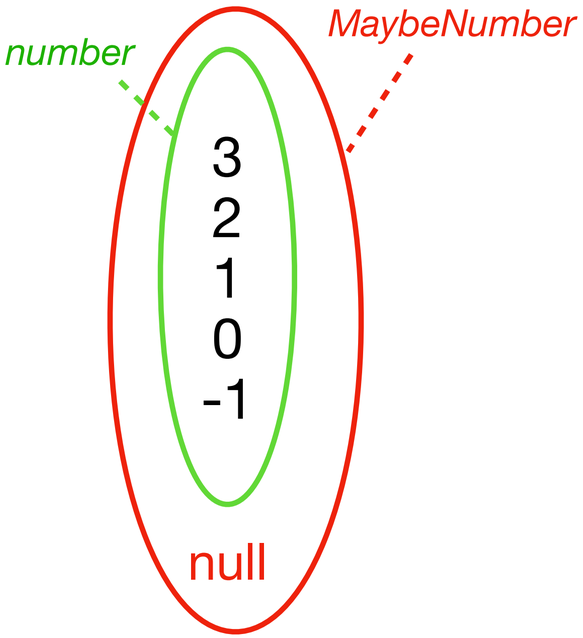
- We are thinking about null value incorrectly.
- Think about the values and it's type, null isn't an element of it type!
- We should exclude null from output types unless we explicitly say they have the null value.
- Let's call number with nullable value -> MaybeNumber



จำนวนสมาชิกใน set

Cardinality

MaybeNumber has integer and null as members, this called cadinarity



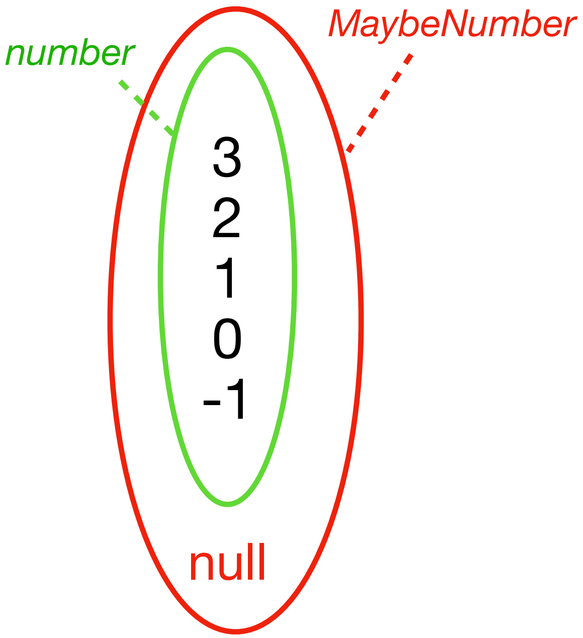
null, 1 means another one value

```
|MaybeNumber| = |number| + 1 type
```

- MaybeInteger is the **cardinality** of the Integer type plus 1
- **Cardinality** of a type, is a number of elements that type has.
- **Cardinality** of a type A represented as |A|

Maybe == Option Type in FP

Let's call Maybe as an Option type




```
type MaybeNumber = number | null  // <- what we have

type Option<A> = A | null         // <- define Optional of
                                     any type

type MaybeNumber = Option<number> // <- now, we can define a
                                     concrete type that reuse the
                                     generic Option type.
```

Some & None

Some and None are 2 variants type of the Option type

```
type Option<A> = Some<A> | None
```

```
interface Some<A> {
  readonly _tag: 'some'
  readonly value: A
}
```

```
const some = <A>(a: A): Option<A> => ({
  _tag: 'some',
  value:  a
})
```

```
interface None {
  readonly _tag: 'none'
}
```

```
const none: Option<never> = {
  _tag: 'none'
}
```

Partial Function

Let's go back and talk about compose Partial Functions

```
function divide2(x: number): number {
  return 2 / x
}

function increment(x: number): number {
  return x + 1
}

const chopBump = compose(increment, divide2)          // compile error

const chopBump = compose(
    x => isNone(x) ? none : some(increment(x.value)),  // chopBump(0) === none,
    divide2                                            // chopBump(8) === some(1.25)
)
```

There is another better way to solve this called Functor.
We'll talk about this later.

Lab : Understand Option Type

1/3 - Define Option, Some, None

```ts
// option.ts

type Option<A> = Some<A> | None
type Some<A> = Readonly<{
  _tag: 'Some',
  value: A
}>
type None = Readonly<{
  _tag: 'None',
}>

export const some = <A>(value: A): Some<A> => ({
  _tag: 'Some',
  value
})

export const none: None = {
  _tag: 'None',
}

export const isSome = <A>(option: Option<A>): option is Some<A> =>
  option._tag === 'Some'

export const isNone = <A>(option: Option<A>): option is None =>
  option._tag === 'None'
```

```ts
// option.test.ts

import { isNone, isSome, none, some } from "./option"

describe('Option', () => {
  it('verify some option', () => {
    expect(isSome(some(1))).toBeTruthy()
    expect(isNone(some(1))).toBeFalsy()
  })

  it('verify none option', () => {
    expect(isNone(none)).toBeTruthy()
    expect(isSome(none)).toBeFalsy()
  })
})
```

Lab : Understand Option Type

2/3 - Use Option Type

```
// option.compose.ts

import { pipe } from 'effect'
import { Option, some, none, isSome } from './option'

const divideTwo = (n: number): number => 2 / n
const liftUp = (n: number): Option<number> => n === 0 ? none : some(n)
const shouldDivideTwo = (o: Option<number>) => isSome(o) ? some(divideTwo(o.value)) : none

describe('Option Compose', () => {
  it('should compose two functions', () => {
    expect(
      pipe(
        1, liftUp, shouldDivideTwo
      )
    ).toEqual(some(2))
  })

  it('should return none if the input is 0', () => {
    expect(
      pipe(
        0, liftUp, shouldDivideTwo
      )
    ).toEqual(none)
  })
})
```

Lab : Effect Option

3/3 - Use Option Type Provided from Effect

```
// option.effect.test.ts

import { pipe } from 'effect'
import { Option, some, none, isSome } from 'effect/Option'

const divideTwo = (n: number): number => 2 / n
const liftUp = (n: number): Option<number> => n === 0 ? none() : some(n)
const shouldDivideTwo = (o: Option<number>) => isSome(o) ? some(divideTwo(o.value)) : none

describe('Option Effect', () => {
  it('should compose two functions', () => {
    expect(
      pipe(
        1, liftUp, shouldDivideTwo
      )
    ).toEqual(some(2))
  })

  it('should return none if the input is 0', () => {
    expect(
      pipe(
        0, liftUp, shouldDivideTwo
      )
    ).toEqual(none)
  })
})
```

Either

Preserve the error message that missing in Option

Either

can result either success or failure

- Option provide info that function may return ***some*** or none ***value***.
- If you need to include additional information, such as an error message, to explain why a value is missing, Either is the preferred choice.

```
type     Option<A> = None     | Some<A>

type Either<E, A> = Left<E> | Right<A>
```

Either

can result either success or failure

- The function should be Total Function. In the other hand, it should always “**Right**”.
- Otherwise, it should be “**Left**” off.

```
type Either<E, A>
```

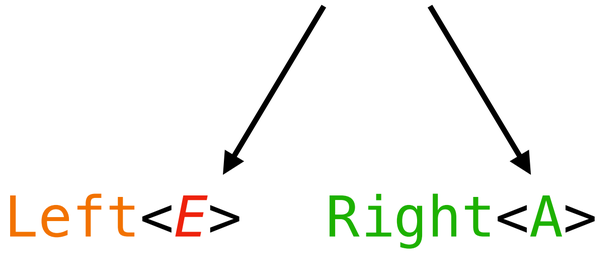
```
Left<E>   Right<A>
```

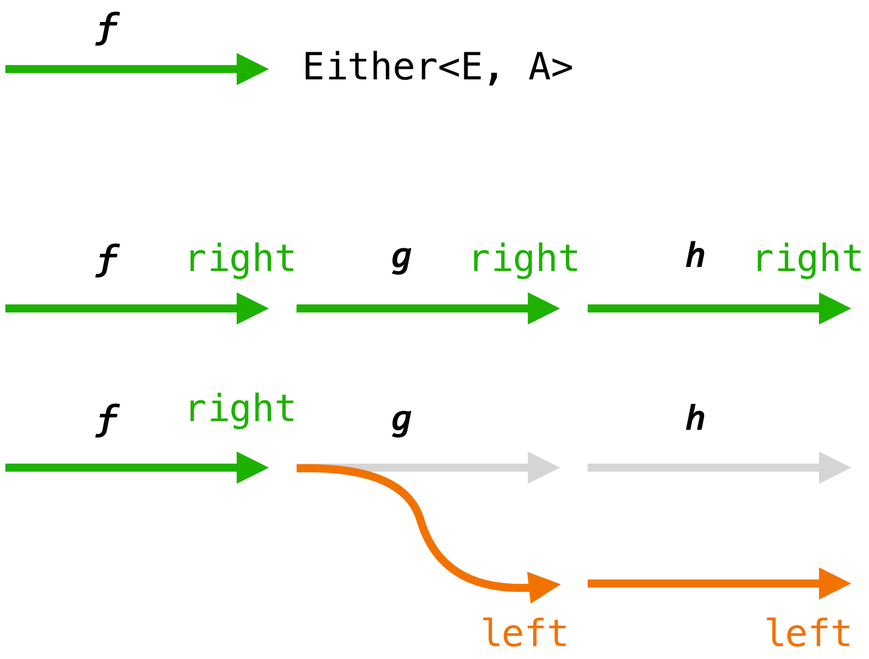
Either

Compose either functions (Railway-Oriented Programming)

- Rather than handle error with try catch, Functional Programing use left or right result to handle error.

```
type Either<E, A>
```





Value or Error

this code is a Partial function we want to avoid

```
function DivideTwoIfEven(num: number): number {
  if (num === 0) {
    throw 'Cannot divide by zero'
  }
  else if (num % 2 == 0) {
    throw 'num is not even'
  }
  else {
    return 2 / num
  }
}
```

Option

Option not provide reason

```
function DivideTwoIfEven(num: number): Option<number> {
  if (num === 0) {
    return none
  }
  if (num % 2 == 0) {
    return none
  }
  return some(2 / num)
}
```

We want the type that can return error to provide the error reason.

Either Type

Type declaration

```
type Either<E, A> = Left<E> | Right<A>
```

Either Type

Implementation. Now, result can be left and right

```
type Either<E, A> = Left<E> | Right<A>

interface Left<E> {
  readonly _tag: 'left'
  readonly left: E
}
interface Right<A> {
  readonly _tag: 'right'
  readonly right: A
}

let result: Either<string, number>
result = { _tag: 'right', right: 12 }
result = { _tag: ‘left’,  left: 'Error' }
```

Either Type

Add helper function to left up any type to Either Left or Right

```
type Either<E, A> = Left<E> | Right<A>

interface Left<E> {
  readonly _tag: 'left'
  readonly left: E
}
interface Right<A> {
  readonly _tag: 'right'
  readonly right: A
}

let result: Either<string, number>
result = { _tag: 'right', right: 12 }
result = { _tag: ‘left’,  left: 'Error' }
```

```
const left = <E, A=never>(e: E): Either<E, A> => ({
    _tag: 'left', left: e
})

const right = <A, E=never>(a: A): Either<E, A> =>
({
    _tag: 'right', right: a
})
```

Either

Use Either to return none with error message

```
function DivideTwoIfEven(num: number): Either<string, number> {
  if (num === 0) {
    return left('Cannot divide by zero')
  }
  if (num % 2 !== 0) {
    return left('num is not even')
  }
  return right(2 / num)
}
```

Lab : Either

1/3 - Define Either, Left, Right

```
// either.ts

export type Either<E, A> = Left<E> | Right<A>

export type Left<E> = Readonly<{
  _tag: 'Left',
  value: E
}>

export type Right<A> = Readonly<{
  _tag: 'Right',
  value: A
}>

export const left = <E>(value: E): Left<E> => ({
  _tag: 'Left',
  value
})

export const right = <A>(value: A): Right<A> => ({
  _tag: 'Right',
  value
})

export const isLeft = <E, A>(either: Either<E, A>): either is Left<E> =>
  either._tag === 'Left'

export const isRight = <E, A>(either: Either<E, A>): either is Right<A> =>
  either._tag === 'Right'
```

Lab : Either

2/3 - Using Either

```
// divideTwo.test.ts

import { Either, left, right, isLeft, isRight } from "./either"

const divideTwo = (divisor: number): Either<string, number> => {
  if (divisor === 0) {
    return left('Division by zero error.')
  } else {
    return right(2 / divisor)
  }
}

describe('divideTwo', () => {
  it('should be Right when divisor is not zero, otherwise should be Left', () => {
    expect(isRight(divideTwo(4))).toBeTruthy
    expect(isLeft(divideTwo(0))).toBeTruthy
  })

  it('should return the result when divisor is not zero', () => {
    const result = divideTwo(4)
    if (isRight(result)) {
      expect(result.value).toBe(0.5)
    } else {
      fail('should be Right')
    }
  })
})
```

Lab : Either

3/3 - Using Either with Effect

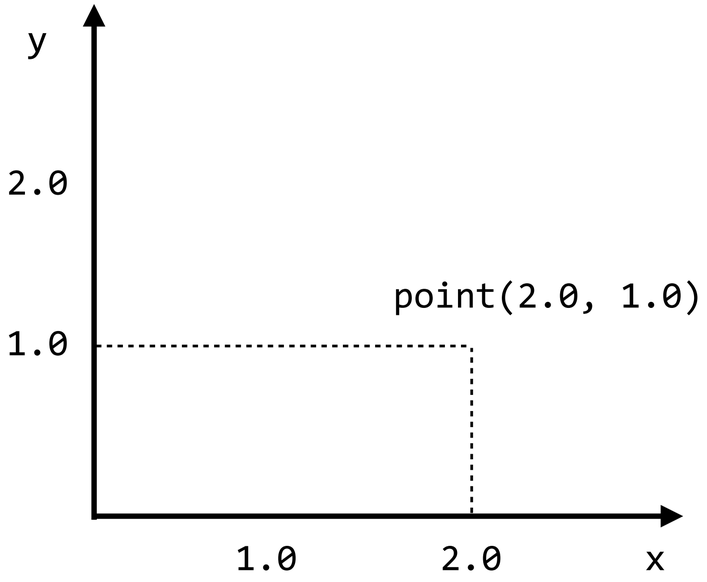
```
// divideTwo-effect.test.ts

import * as Either from "effect/Either"

const divideTwo = (divisor: number): Either.Either<number, string> =>
  divisor === 0
    ? Either.left("Division by zero error.")
    : Either.right(2 / divisor)

describe('divideTwo Effect', () => {
  it('should be Right when divisor is not zero, otherwise should be Left', () => {
    expect(
      divideTwo(4)
    ).toEqual(Either.right(0.5))

    expect(
      divideTwo(0)
    ).toEqual(Either.left("Division by zero error."))
  })

  it('should return the result when divisor is not zero', () => {
    const result = divideTwo(4)
    if (Either.isRight(result)) {
      expect(
        result.right
      ).toBe(0.5)
    } else {
      fail('should be Right')
    }
  })
})
```

ADT & Pattern Matching

Extract value with Pattern Matching

Compound Data Type

Group data objects into Tuples



Point in 2D plane

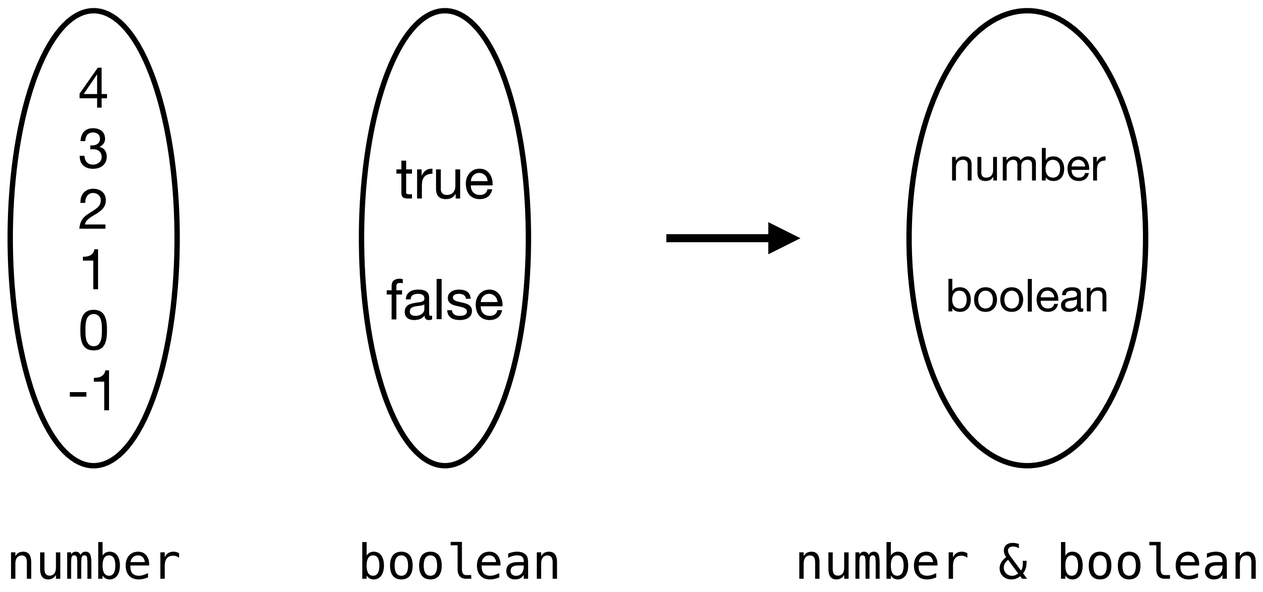
- Primitive data types are useful but often nor sufficient to modeling the real world phenomena.
- For example :- to represent the coordinates in 2D graphics. We need two numbers to represent x, y coordinator.
- We need to glue those x and y together into a compound data object as a single concept.
- In many programming languages support data type object to glue 2 or more data types together. It's called ***Tuple***.

```
const point: [number, number] = [1, 2]
const [x, y] = point
```

Composing Types

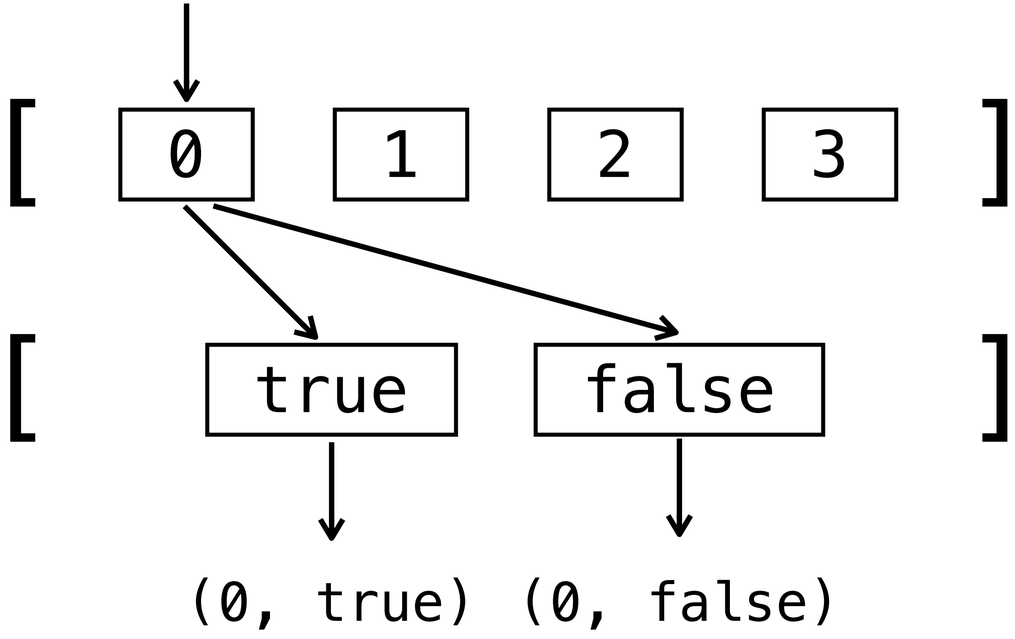
let's call it type

composing "set of values" with another "set of values" to create a "new set of values".



Composing Types

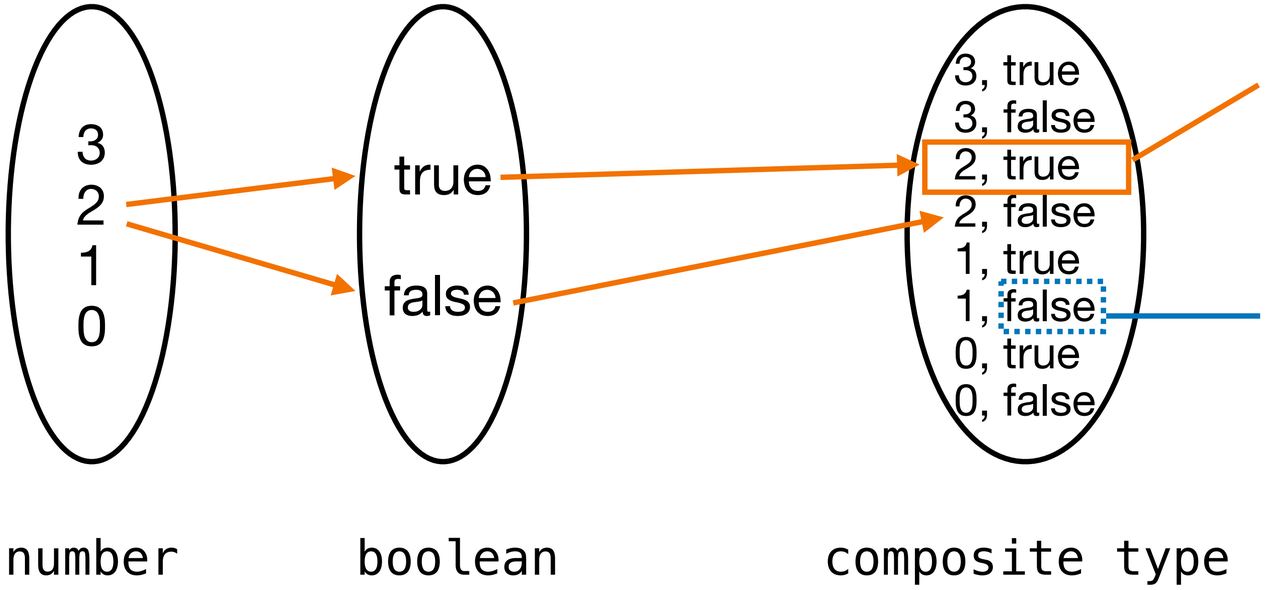
Product Type - Think list as possibility




```
[(0, true), (0, false), (1, true), ... (3, true), (3, false)]
```

Composing Types

Product Type & Field



The resulting composite type is called *Product Type.*

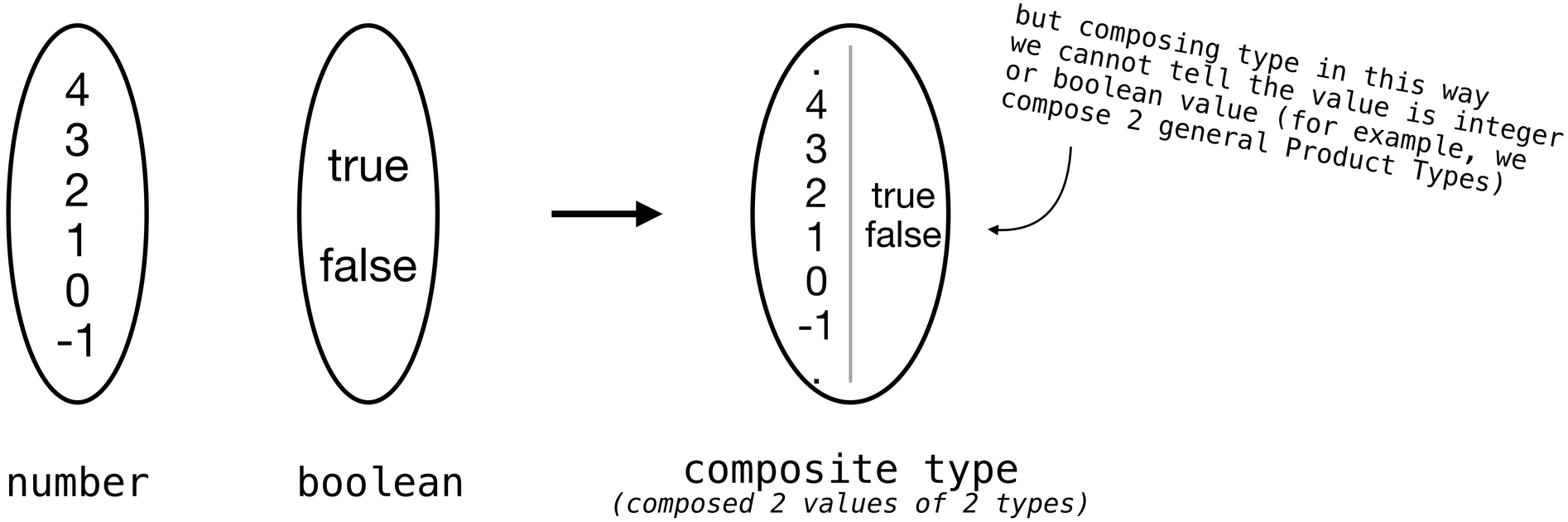
In composite type, each value in side the pair is called a field in our new type.

Or called cardinality Type.

```
[(0, true), (0, false), (1, true), ... (3, true), (3, false)]
```

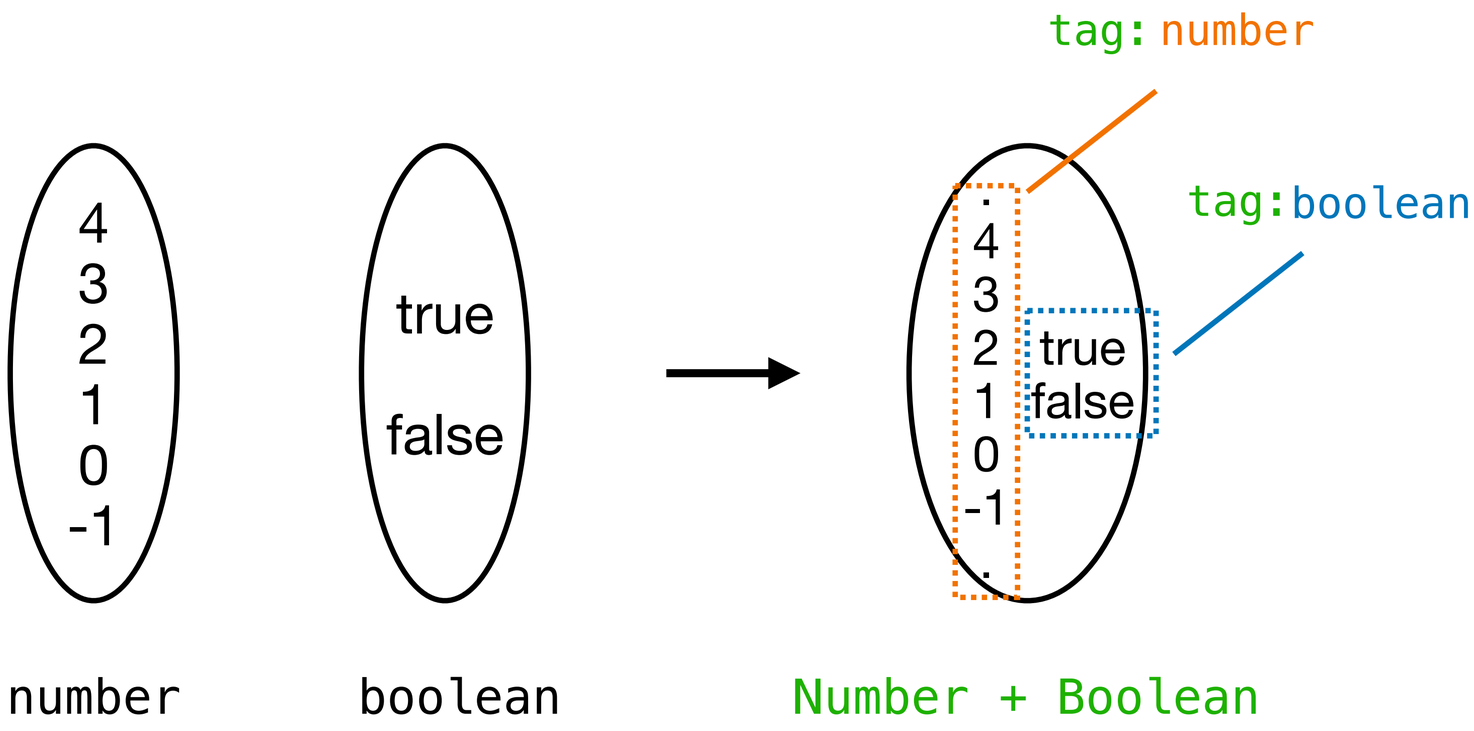
Composing Types

Putting every values from both type next to each other in our composite type.



Composing Types

Grouping and tagging the values in Composite Type



The composite type that compose like this called Sum Type.

ADT

- ADT stands for Algebraic Data Type.
- ADT is basically a composite type using operations.
  - Product operation
  - Sum operation
- We want to use composite type to store data inside them.
- The operation that work with composite type to extract data from it called *Pattern Matching*.

Algebraic Data Types

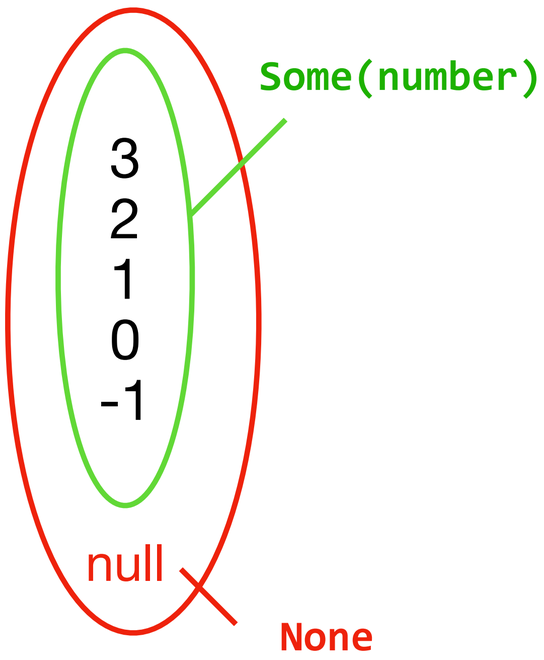
Example :-

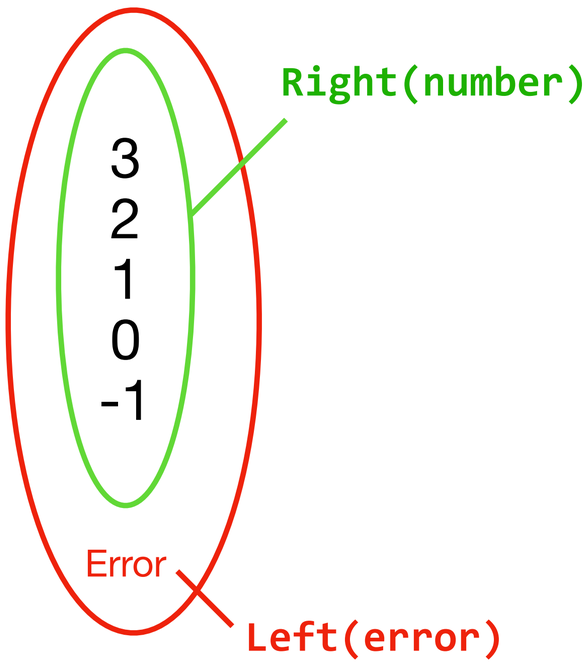
```
type DayOfWeek = 'Monday'
               | 'Tuesday'
               | 'Wednesday'
               | 'Thursday'
               | 'Friday'
               | 'Saturday'
               | 'Sunday'
```

```
type Shape = Circle
           | Square
           | Rectangle
           | Triangle

interface Circle {}
interface Square {}
interface Rectangle {}
interface Triangle {}
```

Option & Either

We've already seen ADT.






```
Option = Some<A> | None
```

```
Either = Left<E> | Right<A>
```

Model Sum Type in TypeScript

Unit Type or Either

sum of number and boolean

```
type NumberOrBool = number | boolean

const x: NumberOfBool = 5
const y: NumberOfBool = true
```

```
type Either<E, A> =
  | { _tag: 'left', left: E }
  | { _tag: 'right', right: A}
```

Either is a generic sum type that can have a left object type or right object type.

```
const x: Either<string, number[]> = { _tag: `left`, left: `Error` }
const y: Either<string, number[]> = { _tag: `right`, left: [1, 2, 3] }
```

Convert composition type to other type

Let's see option type

2 variants, none & some<boolean>

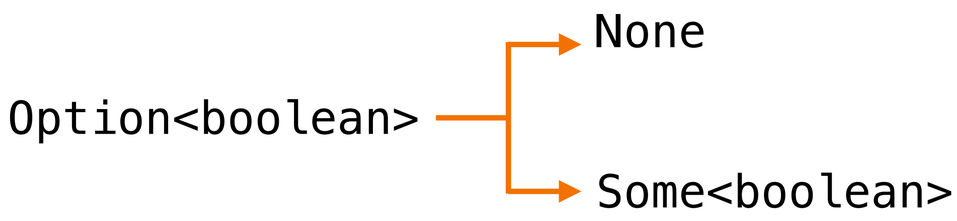
```
type Option<boolean> = None | Some<boolean>
```

we want to extract value from 2 variants

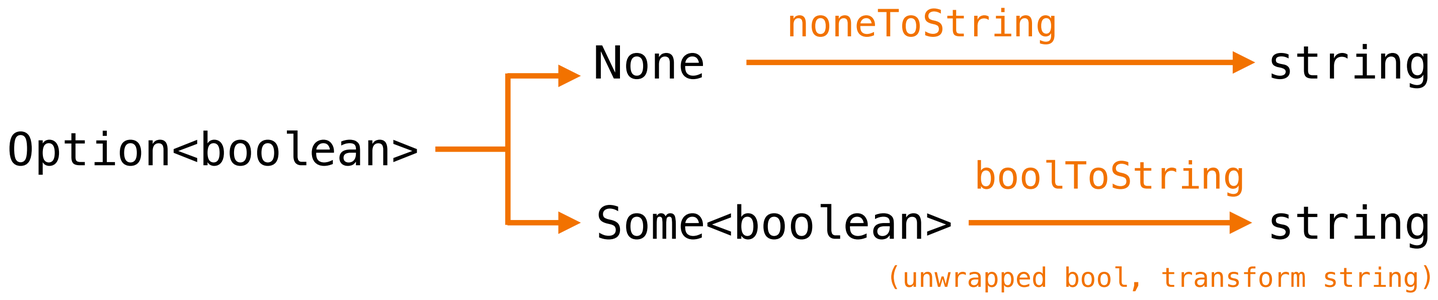
```
Option<boolean> --toString--> string
```

Pattern Matching

Extract and transform, from Option<boolean> to string.



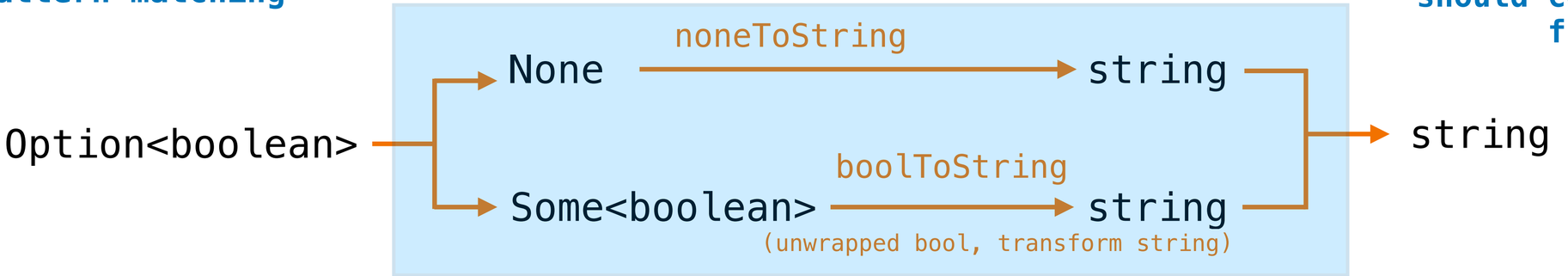
we look at each variant in option (composite type) case by case then extract data from each variant and transform all of the result to a desired value type.



Because when implement pattern matching un FP, FP don't have partial function.

That's means, Pattern matching should cover all the values from its input type.

This called, pattern matching



Lab 1 : Pattern Matching - Option

1/3 - Option

```
// Option ------------------------

type Option<A> =
  | None
  | Some<A>

type None = Readonly<{
  _tag: 'none'
}>

type Some<A> = Readonly<{
  _tag: 'some'
  value: A
}>

const some = <A>(x: A): Option<A> => ({ _tag: 'some', value: x })
const none: Option<never> = { _tag: 'none' }
const isNone = <A>(x: Option<A>): x is None => x._tag === 'none'

export { Option, some, none }
```

```
interface None {
  readonly _tag: 'none'
}

interface Some<A> {
  readonly _tag: 'some'
  readonly value: A
}
```

Lab 1 : Option

2/3 - Option Type Pattern Matching

```
// Pattern Matching -------------------------

type Match = <A, B>(
  onNone: () => B,
  onSome: (a: A) => B
) =>
  (x: Option<A>) => B

const match: Match = (onNone, onSome) => x =>
  isNone(x) ? onNone() : onSome(x.value);

export { Option, some, none, match }
```

```
// Test------------------

import { match, Option, some } from "./adt-option"

describe('Pattern Matching with Option Type', () => {
  it('should match Option type to string', () => {

    const mayBeNum: Option<number> = some(10)
    const matched = match(
      () => `number doesn't exist`,
      (x) => `${x}`
    )(mayBeNum)

    expect(matched).toEqual('10')

  })
})
```

Lab 1 : Option

3/3 - MatchW (Widen)

```
...
const match: Match = (onNone, onSome) => x =>
  isNone(x) ? onNone() : onSome(x.value)

type MatchW = <A, B, C>(
  onNone: () => B,
  onSome: (a: A) => C
) =>
  (x: Option<A>) => B | C

const matchW: MatchW = (onNone, onSome) => x =>
  isNone(x) ? onNone() : onSome(x.value)

export { Option, some, none, match, matchW }
```

```
it('should match widen Option type to number when it is some type', () => {
  const mayBeNum: Option<number> = some(10)
  const matched = matchW(
    () => `number doesn't exist`,
    (x) => x
  )(mayBeNum)
  expect(matched).toEqual(10)
})
```

```
it('should match widen Option type to string when it is none type', () => {
  const mayBeNum: Option<number> = none
  const matched = matchW(
    () => `number doesn't exist`,
    (x) => x
  )(mayBeNum)
  expect(matched).toEqual(`number doesn't exist`)
})
```

Lab 2 : Pattern Matching - Either

1/2 - Either

```
// Either -------------------------

type Either<E, A> = Left<E> | Right<A>

type Left<E> = Readonly<{
  _tag: 'left'
  left: E
}>

type Right<A> = Readonly<{
  _tag: 'right'
  right: A
}>

const left = <E, A = never>(e: E): Either<E, A> => ({
  _tag: 'left', left: e
})

const right = <A, E = never > (a: A): Either<E, A> => ({
  _tag: 'right', right: a
})

const isLeft = <E, A>(x: Either<E, A>): x is Left<E> =>
  x._tag === 'left'

export { Either, left, right, isLeft }
```

Lab 2 : Pattern Matching - Either

2/2 - Pattern Matching Either

```
// Pattern Matching -------------------------

type Match = <E, A, B>(onLeft: (e: E) => B, onRight: (a: A) => B) =>
  (x: Either<E, A>) => B

const match: Match = (onLeft, onRight) => x =>
  isLeft(x) ? onLeft(x.left) : onRight(x.right)

export { Either, left, right, isLeft, match }
```

```
it('should match right to string', () => {
  const errorOrNum: Either<string, number> = right(12)
  expect
    (match(
      (e: string) => e,
      (a: number) => `value is ${a}`
    )(errorOrNum)
  ).toEqual(`value is 12`)
})
```

```
it('should also match left to string', () => {
  const errorOrNum: Either<string, number> = left(`error`)
  expect(
    match(
      (e: string) => e,
      (a: number) => `value is ${a}`
    )(errorOrNum)
  ).toEqual(`error`)
})
```

Lab 3: Pattern Matching - Effect

1/2 - Match Option with Effect

```
import { Option, pipe } from "effect"

const matchOption = (option: Option.Option<number>): string =>
  pipe(
    option,
    Option.match({
      onNone: () => 'none',
      onSome: (x: number) => `${x}`
    })
  )

describe('Match Effect', () => {
  it('should match Option success', () => {
    expect(
      matchOption(Option.some(1))
    ).toBe('1')
  })

  it('should match Option none', () => {
    expect(
      matchOption(Option.none())
    ).toBe('none')
  })
})
```

Lab 3: Pattern Matching - Effect

2/2 - Match Option with Effect

```
import { Option, pipe } from "effect"

const matchEither = (either: Either.Either<number, string>): string =>
  pipe(
    either,
    Either.match({
      onLeft: (e: string) => e,
      onRight: (x: number) => `${x}`
    })
  )

describe('Match Effect', () => {
  it('should match Either success', () => {
    expect(
      matchEither(Either.right(1))
    ).toBe('1')
  })

  it('should match Either left', () => {
    expect(
      matchEither(Either.left('error'))
    ).toBe('error')
  })

  it('should match Either right', () => {
    expect(
      matchEither(Either.right(1))
    ).toBe('1')
  })
})
```

[x] Lab 4 : Pattern Matching - List

1/3 - List (optional lab)

```
// List ----------------------------------------

type List<A> = Nil | Cons<A>

type Nil = Readonly<{
  _tag: 'nil'
}>

type Cons<A> = Readonly<{
  _tag: 'cons'
  head: A
  tail: List<A>
}>

const nil: List<never> = { _tag: 'nil' }
const cons = <A>(head: A, tail: List<A>): List<A> => ({
  _tag: ‘cons', head, tail
})
```

```
const isNil = <A>(xs: List<A>): xs is Nil => xs._tag === 'nil'

const showList = <A>(xs: List<A>): string =>
  isNil(xs)
    ? ''
    : `${xs.head}` +
      `${isNil(xs.tail) ? "" : ", "}` +
      `${showList(xs.tail)}`


// Pattern Matching ----------------------------------------

// Start hete…



export { Cons, cons, nil, showList }
```

[x] Lab 4 : Pattern Matching - List

2/3 - List Pattern Matching

```
// Pattern Matching -------------------------------

type Match = <A, B>(
  onNil: () => B,
  onCons: (xs: List<A>) => B
) => (xs: List<A>) => B

const match: Match = (onNil, onCons) => xs =>
  isNil(xs) ? onNil() : onCons(xs)

export { Cons, cons, nil, showList, match }
```

```
import {
  Cons, cons, match, nil, showList
} from ‘./adt-list'

describe('Pattern Matching with Option Type', () => {
  it('should match list to string', () => {
    const list = cons(1, cons(2, cons(3, cons(4, nil))))
    expect(
      match(
        () => '',
        (xs) => showList(xs)
      )(list)
    ).toEqual('1, 2, 3, 4')
  })

  it('should match null list to empty string', () => {
    expect(
      match(
        () => '',
        (xs) => showList(xs)
      )(nil)
    ).toEqual('')
  })
})
```

[x] Lab 4 : Pattern Matching - List

3/3 - List Pattern Matching with ts-pattern

```
import {
  Cons, cons, match, nil, showList
} from './adt-list'

import { match as matchFp} from 'ts-pattern'

describe('Pattern Matching with Option Type', () => {

// ...

  it('should match list to string with ts-pattern', () => {
    const list = cons(1, cons(2, cons(3, cons(4, nil))))
    expect(
      matchFp(list)
        .with({ _tag: 'nil' }, () => '')
        .with({ _tag: 'cons' }, (xs) => showList(xs))
        .exhaustive()
    ).toEqual('1, 2, 3, 4')
  })
})
```

Higher-Kinded Types

Type that accept another type and return a new Type

Let’s Recap

from earlier lesson

- We learn that values grouped by types.
- We know the benefit about coding by dealing with types rather than values.
- Functions are simply mapping between different types.
- Higher-order function that can pass function and return function.
- Next, we will learn about another abstraction above types and type constructors.

The Value and Type Universe

start by looking at values again, then type, and something above of its.

Types made compiler help us
to prevent error
when dealing with different values.

Value Universe	Type Universe	???
12	number	There are another universe that abstract the similar patterns in the Type Universe.
`hello`	string	
true	number => boolean	
	Option<boolean>	

Option

Think about Option

- **Option** itself seem to be incomplete. We need to pass a concrete type to it's type constructor.

```
Option<???>  ->  Option<boolean>

Option: Type ->  Type
```

boolean

Option<boolean>

- It's acting like functions that accept another type (boolean) to construct itself (option). And this is how we abstract over the Type universe.

- We called the type, like Option, that abstract type in to the higher level as ***Kind***.

Kind

The abstraction level that abstract the Type Universe into the higher level.

Value Universe

```
12

`hello`

true
```

Type Universe

```
number

string

number => boolean

Option<boolean>

string[]
```

Kind Universe

```
Type -> Type

A[]

Option<A>
  -> some<A>
  -> none

(Type -> Type) -> Type

Either<E,A>
  -> Left<E>
  -> Right<A>
```

Kind

the definition

- Kind universe is an ***abstraction*** over the Type universe.
- Kind is a classification for *types* and *type constructors*.
- Type constructors in the Kind universe are like *currying functions* in the Type universe.

```
Either<E, A>    =>     Either -> E -> A
```

- Think about array for example. Array keeps a sequence of items with a same type. It's doesn't care about the items it holds. All the operations are what array care about.

```
Array<Number>    [1, 2, 3, 4]
Array<boolean>   [true, false, true]
Array<string>    [`hello`, `world`, `!`]
```

- In the other word, Array is *abstracting over the type* it holds.

Higher-Kinded Type

look at the Kind closely…

- Looking at the Array again…

```
Array<Number>     [1, 2, 3, 4]
Array<boolean>    [true, false, true]
Array<string>     [`hello`, `world`, `!`]

Array<A>          A is any type
```

- Then, we also have another Kind universe…

```
Array <Number>    [1, 2, 3]
List  <Number>    cons(1, cons(2, cons(3, nil)))
Option<Number>    some(1)
```

- Can we have a generic type constructor that represents the common behaviour between type constructors?

- **The answer is YES. That's called *Higher-Kinded Type*.**

F<Type>

Higher-Kinded Type

A	Type (like Number)
Array<A>	Life up A with Array of A **Type -> Type**
F<A>	F can be Array, List, Option of A. (F is a Higher-kinded Type)
Array<A>	Replace F with Array **Type -> Type**
Array<Number>	**(Type -> Type) -> Type**

Higher-Kinded Type

Each Functor has map function, but the implementation is different

Type -> Type

```
type map_option =      <A, B>(f: (a: A) => B) => (x: Option<A>)     => Option<B>
type map_List   =      <A, B>(f: (a: A) => B) => (x: List<A>)       => List<B>

type map_either =   <E, A, B>(f: (a: A) => B) => (x: Either<E, A>) => Either<E, B>
```

Type -> Type -> Type

Type Class

To simplify it in to Type -> Type

Type -> Type

```
type map_option =      <A, B>(f: (a: A) => B) => (x: Option<A>)    => Option<B>
type map_List   =      <A, B>(f: (a: A) => B) => (x: List<A>)      => List<B>

type map_either =   <E, A, B>(f: (a: A) => B) => (x: Either<E, A>) => Either<E, B>

type map_either<E> =   <A, B>(f: (a: A) => B) => (x: Either<E, A>) => Either<E, B>

Functor<F> {    <- This called Type Class
    map: <A, B>(f: A => B) => F<A> => F<B>
}
```

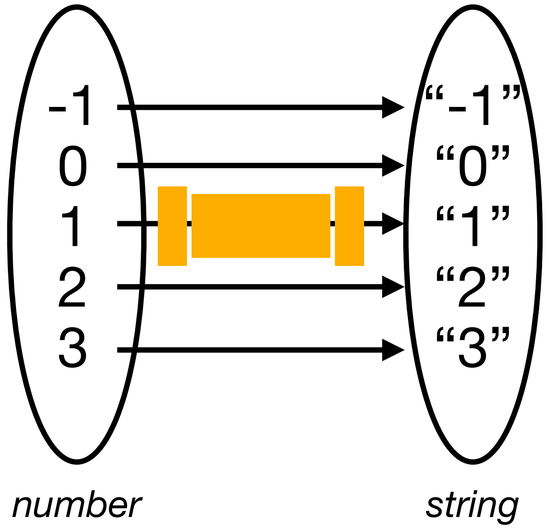
We'll talk about Type-Class in the next chapter.

IO

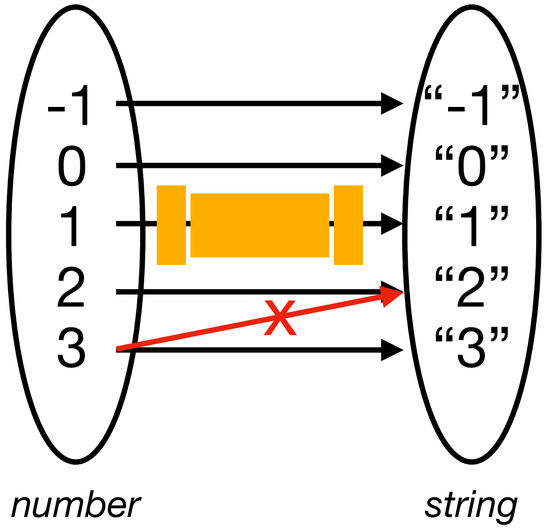
Compose synchronous computations with Side Effects

Pure Functions

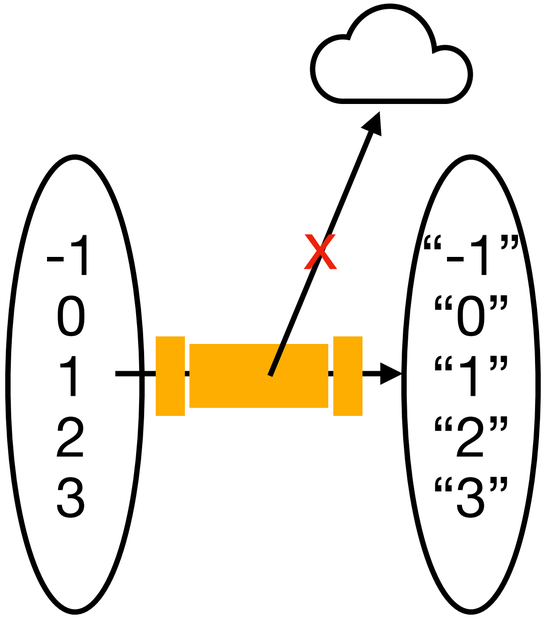
revisit function definition



Total Function



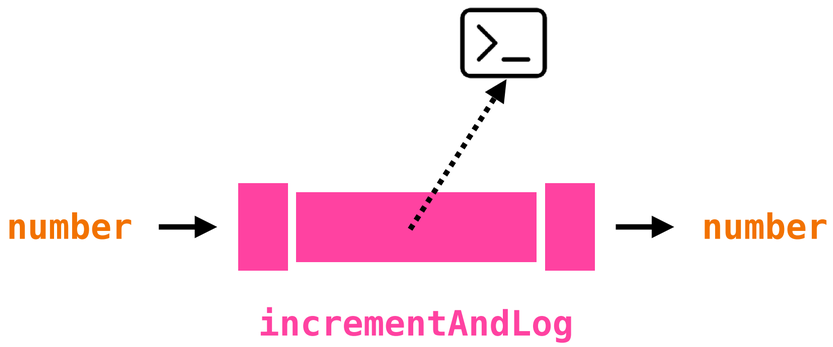
Deterministic



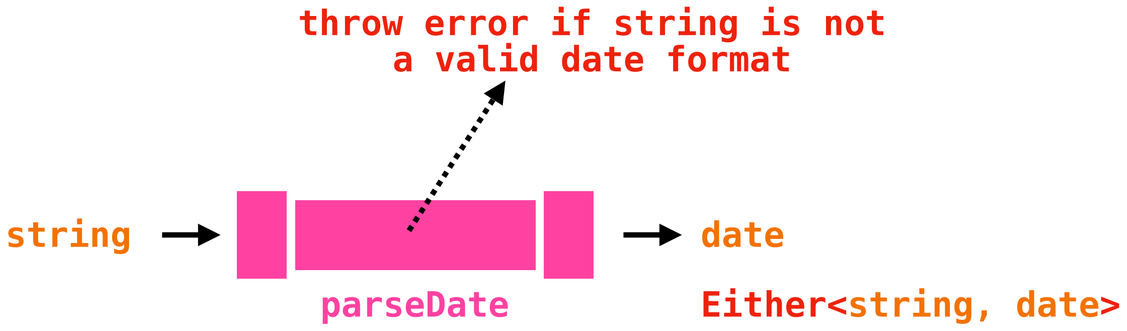
No Side effects

Side Effects

it implicitly dependent on the context



write a message to console that may not be able to compose



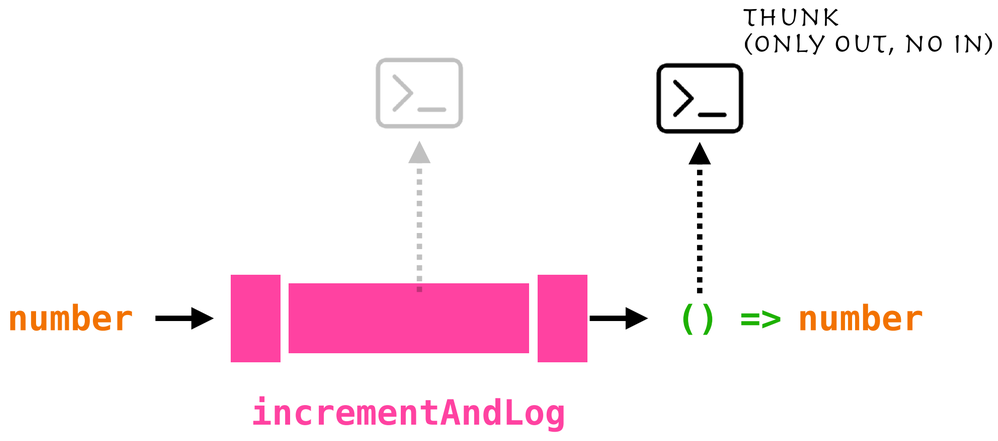
however, think about the error that happened when compose functions.

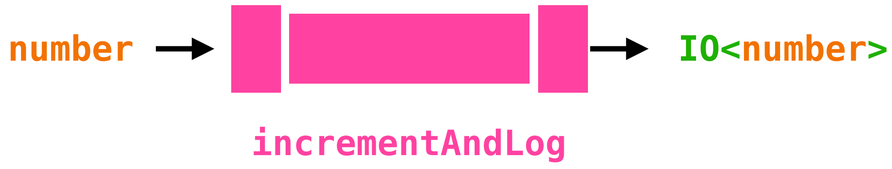
We've already solved this problem with Either.

We'll use this idea to model any kinds of Side Effects

Thunk

a pure function that has no input, handle side effect, and always return the same result.





- We can add small function that accept no argument that extract the side effect part into a new small function.
- We called this function “**Thunk**”.
- This make “`incrementAndLog`” pure.
- To modeling side effects, we came up with a new structure specifically to handle side effect. Let’s named it - IO.

IO

```
type IO<A> = () => A
```

- **IO** model computational

  IO wrap generic type constructor over any type ‘A’. It’s a computational wrapped in a box so that we can pass it around as data. And return value based on type it wrapped. This function doesn’t receive and input and always return a value of type ‘A’.

- **IO** has side effects

  It can get or set values outside of its body.

- **IO** has no exception

  When execute IO. It’s doesn’t error out.

- **IO** is synchronous

  It should be synchronous. When we execute the IO, it will block the execution until it’s done.

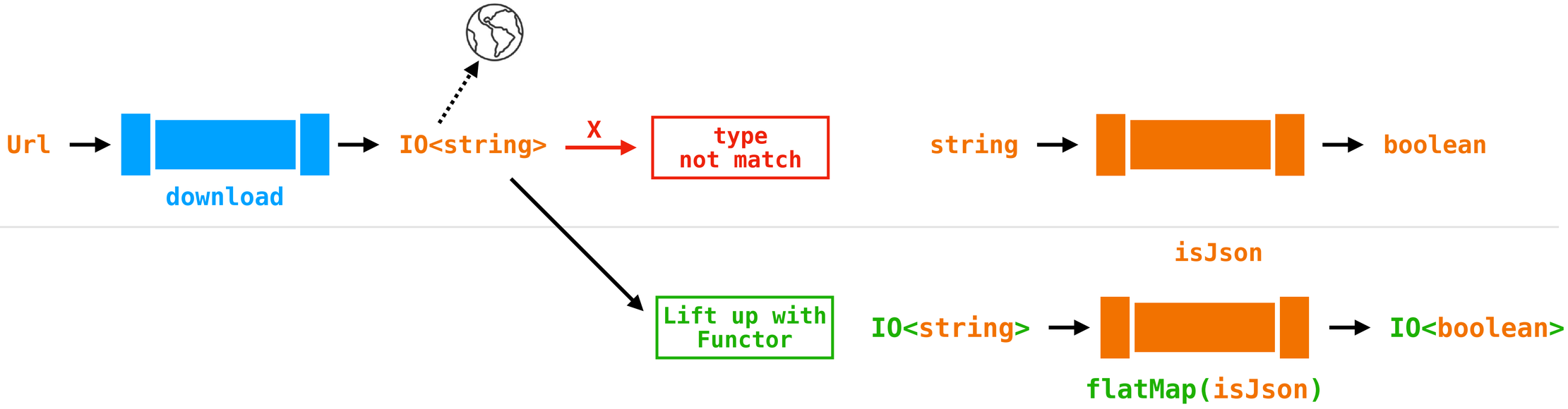
IO

example

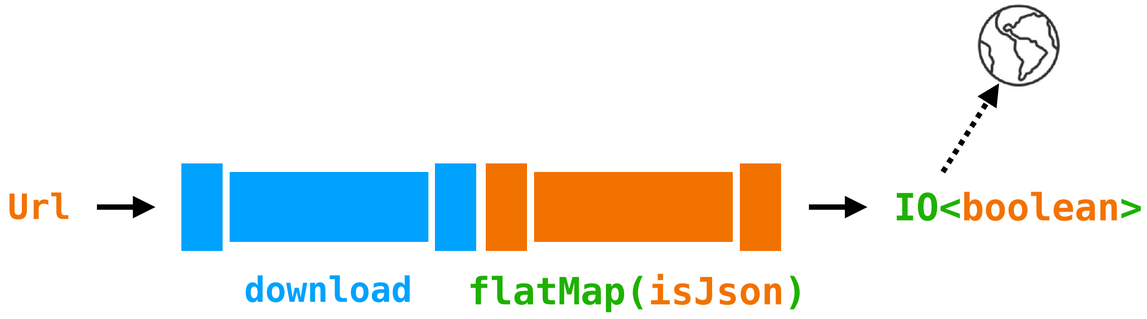
```
const increment = (value: number): number => {
  const output = value + 1
  console.log(output)
  return output
}

increment(5)

// ==========================================================

type IO<A> = () => A

const ioIncrement = (value: number): IO<number> => {
  const output = value + 1
  return () => {
    console.log(output)
    return output
  }
}

const incrementAndLog = ioIncrement(5)
incrementAndLog()
```

Composing IO

Lift up with functor to map string (input) to IO context

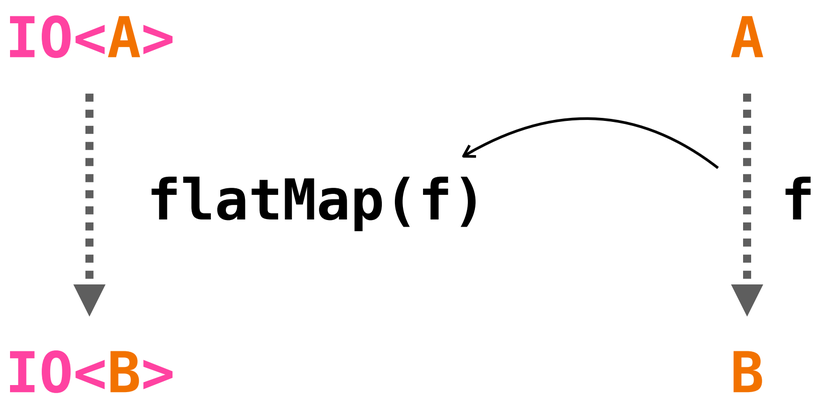


Now we can compose 2 pure/ reliable functions together and delay side effect handling to the edge of the application.



IO as a Functor

IO that can be map from type to type



IO with Error

Handling error with Either - IOEither

```
IO<Either<E, A>>        IOEither<A>
        ⋮                    ⋮
   flatMap(f)           flatMap(f)
        ▼                    ▼
IO<Either<E, B>>        IOEither<B>
```

Since IO and Either are Functor, we can compose then into a new Functor.

IO Summary

IO is not a function - it's a structure that holds data

- **IO** Provide us a tool to handle ***side effects***.
- **IO** is a covariant ***Functor***.
- **IO** does not error out and always provide its wrapped type.
- **IO<Either<E, A>** can be composed to create **IOEither<E, A>**.
- **IO** delays running ***Synchronous*** side effect to a later time.
- **IO<Promise<A>>** can be used to model ***Asynchronous*** side effect.
- **IO<Promise<A>>** can be composed to create **Task<A>**.

Lab 1 : IO

```
import { Effect, Console } from "effect"
import { pipe } from "effect/Function"

const readLine: Effect.Effect<string> = Effect.succeed("OK")

const printLine = (message: string): Effect.Effect<void> =>
  Console.log(message)

const program = pipe(
  readLine,
  Effect.map(str => `You entered: ${str}`),
  Effect.tap(message => printLine(message))
)
```

```
// io.test
import { Effect } from "effect";
import { program } from "./io";

describe("io", () => {
  it("should be able to console log", () => {
    const result = Effect.runSync(program);
    expect(result).toEqual('You entered: OK');
  });
});
```

Lab 2 : IOEither

```ts
import { Effect, Console } from "effect"
import { right } from "effect/Either"
import { pipe } from "effect/Function"

// In this lab. IOEither wrap Effect because we want Error on the `left`, and right on the `right`
export type IOEither<E, A> = Effect.Effect<A, E>

const printLine = (message: string): IOEither<never, void> => Console.log(message)
const printHello: IOEither<never, void> = printLine('Hello')

export const program = (readLine: IOEither<string, string>) =>
  pipe(
    printHello,
    Effect.flatMap(() => readLine),
    Effect.flatMap((str) => right(`You entered: ${str}`)),
    Effect.match({
      onFailure: () => 'Error',
      onSuccess: (message) => message
    })
  )
```

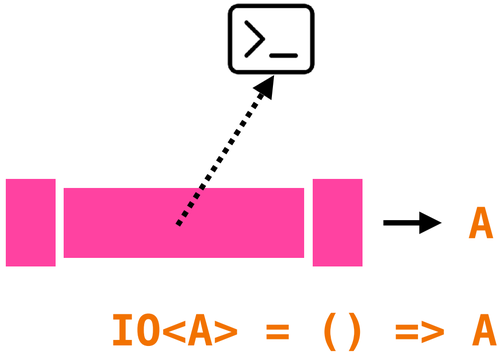
```ts
import { Effect } from "effect"
import { IOEither, program } from "./ioEither"

describe("ioEither", () => {
  const rightReadLine: IOEither<string, string> = Effect.succeed('OK')
  const leftReadLine: IOEither<string, string> = Effect.fail('Error')

  it("should be able to read string from console and print it", () => {
    expect(
      Effect.runSync(program(rightReadLine))
    ).toEqual('You entered: OK')
  })

  it("should be able to read string from console", () => {
    expect(Effect.runSync(program(leftReadLine))).toEqual('Error')
  })
})
```

Task

Compose asynchronous computations with Side Effects

Revisit IO Structure

```
type IO<A> = () => A
```



- **IO** model computational

  IO wrap generic type constructor over any type ‘A’. It’s a computational wrapped in a box so that we can pass it around as data. And return value based on type it wrapped. This function doesn’t receive and input and always return a value of type ‘A’.

- **IO** has side effects

  It can get or set values outside of its body.

- **IO** has no exception

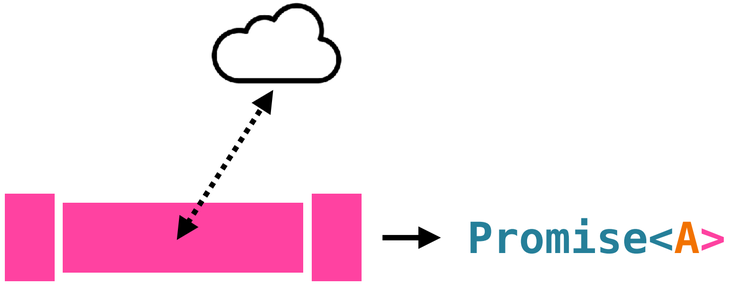
  When execute IO. It’s doesn’t error out.

- **IO** is synchronous

  It should be synchronous. When we execute the IO, it will block the execution until it’s done.

Asynchronous in TypeScript

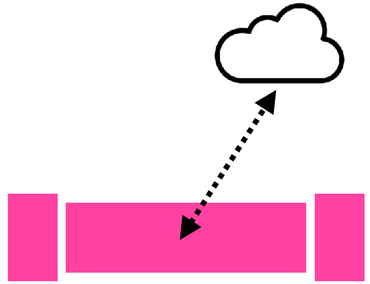
What if we want Asynchronous IO?




```
function fetchBooks(): Promise<Book[]> {
  return fetch("https://jsonplaceholder.typicode.com/posts")
    .then(res => res.json())
    .then(data => data as Book[])
}
```

IO with Asynchronous

What if we want Asynchronous IO?



→ `Promise<A>`

`IO<Promise<A>>`

```
function fetchBooks(): Promise<Book[]> {
  return fetch("https://jsonplaceholder.typicode.com/posts")
    .then(res => res.json())
    .then(data => data as Book[])
}
```

we use Promise to handle asynchronous here and wrap side effect with IO

```
function fetchBooks(): IO<Promise<Book[]>> {
  return () => fetch("https://jsonplaceholder.typicode.com/posts")
    .then(res => res.json())
    .then(data => data as Book[])
}
```

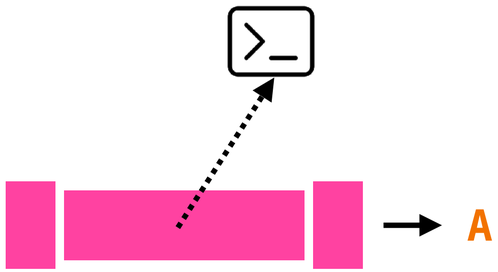
Can we just merge IO and Promise into a new structure? Yes. We call Task.

Task

Handle side effect asynchronously

We can simplify `IO<Promise<A>>` ***with*** *Task*

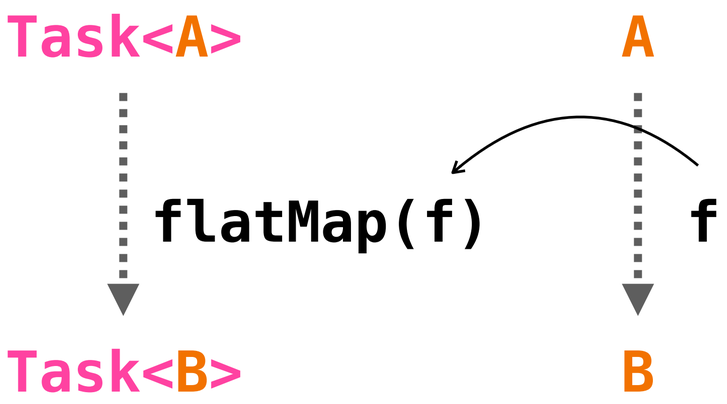
- ***Task*** *model computational*
- ***Task*** *has side effects*
- ***Task*** *has no exceptions*
- ***Task*** *is* ***Asynchronous***




```
Task<A> = () => IO<Promise<A>>
```

Task as a Functor

Task that can be map from type to type



- We can work with Task like IO, Either, Option and all structures we've seen before.
- We apply operations on Task using existing TypeClass operation it supports.
- We can have a map (flatMap) function for Task because Task is a Functor.

Task with Error handling

Task<Either<E, A>>

- By definition, Task should never fail or throw error.
- If you want to handle error due to side effect cannot be trusted. We can use Task to wrap Either.
- Instead of define `Task<Either<E,A>>`. We can define a dedicated structure called `TaskEither<E, A>`.

Lab 1 : TaskEither

fp-ts

```
import { pipe } from "fp-ts/function"
import * as TE from "fp-ts/TaskEither"

type User = {
  id: number
  name: string
  email: string
}

const fetchResponse = (url: string): TE.TaskEither<string, Response> =>
  TE.tryCatch(
    () => fetch(url),
    () => "Network error"
  )

const checkStatus = (res: Response): TE.TaskEither<string, Response> =>
  res.ok
    ? TE.right(res)
    : TE.left(`HTTP error: ${res.status}`)

const parseJson = (res: Response): TE.TaskEither<string, unknown> =>
  TE.tryCatch(
    () => res.json(),
    () => "Invalid JSON"
  )

const fetchUsers = (url: string): TE.TaskEither<string, User[]> =>
  pipe(
    fetchResponse(url),
    TE.flatMap(checkStatus),
    TE.flatMap(parseJson),
    TE.map((data) => data as User[])
  )

export const requestUsers: TE.TaskEither<string, User[]> = pipe(
  fetchUsers("https://jsonplaceholder.typicode.com/users"),
  TE.map((users) => users.map(u => ({ id: u.id, name: u.name, email: u.email }))),
  TE.orElseW((err) => TE.left(err))
)

// requestUsers().then(console.log)
```

Lab 2 : TaskEither

Effect

```
import { Effect } from "effect"

type User = {
  id: number
  name: string
  email: string
}

const fetchResponse = (url: string) =>
  Effect.tryPromise({
    try: () => fetch(url),
    catch: () => "Network error"
  })

const checkStatus = (res: Response) =>
  res.ok
    ? Effect.succeed(res)
    : Effect.fail(`HTTP error: ${res.status}`)

const parseJson = (res: Response) =>
  Effect.tryPromise({
    try: () => res.json(),
    catch: () => "Invalid JSON"
  })

const fetchUsers = (url: string) =>
  fetchResponse(url).pipe(
    Effect.flatMap(checkStatus),
    Effect.flatMap(parseJson),
    Effect.map((data) => data as User[])
  )

export const requestUsers =
  fetchUsers("https://jsonplaceholder.typicode.com/users").pipe(
    Effect.map(users => users.map(u => ({ id: u.id, name: u.name, email: u.email }))),
    Effect.catchAll(err => Effect.succeed({ error: err }))
  )

// Effect.runPromise(requestUsers).then(console.log)
```

Lab 3 : TaskEither (Optional)

Fetch JSON data example

```
import { tryPromise, catchAll, succeed, runPromise } from "effect/Effect"

type CryptoSnapshotItem = {
  coin: string
  priceUsd: number
  fetchedAt: string
}

const DEFAULT_COINS = ["bitcoin", "ethereum", "solana"]
const DEFAULT_VS_CURRENCY = "usd"

const urlWith = (coins: string[]) =>
  (vsCurrency: string): string =>
    `https://api.coingecko.com/api/v3/simple/price?ids=${coins.join(",")}&vs_currencies=${vsCurrency}`

const fetchSnapshot = (url: string) =>
  tryPromise({
    try: () => fetch(url)
      .then(res => res.json() as Promise<Record<string, { usd: number }>>)
      .then(data => Object.entries(data).map(toCryptoSnapshotItem)),
    catch: (error) => String(error),
  })

const toCryptoSnapshotItem = ([coin, { usd }]: [string, { usd: number }]): CryptoSnapshotItem => ({
  coin,
  priceUsd: usd,
  fetchedAt: new Date().toISOString(),
})

fetchSnapshot(urlWith(DEFAULT_COINS)(DEFAULT_VS_CURRENCY))
  .pipe(catchAll(err => succeed({ error: err })))
  .pipe(runPromise)
  .then(console.log)
```

Thank you

See you tomorrow